Jailbreaking Language-Specific LLMs: A Multilingual Security Evaluation of Sabia-3.1

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***Abstract*—Large Language Models (LLMs) are increasingly deployed in multilingual contexts, yet their security robustness across languages remains under-explored. This paper investigates the vulnerability of Sabia-3.1, a Brazilian LLM, to cross-lingual jailbreak attacks. Using an adapted version of the MultiJail dataset, we evaluate 1260 adversarial prompts translated into English, Brazilian Portuguese, Italian, and Swahili. Responses were classified via GPT-4o into safe, unsafe, or invalid, and a Jailbreak Success Rate (JSR) was computed per language. Results show complete resistance to jailbreaks in Portuguese and Italian (JSR = 0.0%), but measurable vulnerabilities in English (JSR = 3.17%) and especially Swahili (JSR = 4.76%). These findings highlight that language models may generalize safety alignment to linguistically similar languages, while remaining susceptible to jailbreaking in both high-resource (EN) and low-resource (SW) languages.**

***Index Terms*—large language model, llm, jailbreaking, multilingual.**

I. Introduction

Large Language Models (LLMs) have become central to modern AI applications, demonstrating strong generalization capabilities across tasks such as content generation, translation, summarization, and question answering. However, as their adoption grows, so do concerns regarding their misuse, such as prompt injection and jailbreak attacks, where users manipulate input prompts to elicit restricted or harmful outputs [1], [2].

While the study of LLM vulnerabilities has expanded rapidly in English-dominant contexts, little attention has been given to models trained primarily in other languages. This gap is significant given that many countries and communities now build and deploy language models adapted to local linguistic and cultural data.

In this study, we investigate the multilingual robustness of Sabia-3.1, a large language model primarily trained on Brazilian Portuguese content [3]. Our focus is on evaluating its vulnerability to translated and cross-lingual jailbreak prompts across four languages with varying degrees of availability: English (EN), Brazilian Portuguese (PT-BR), Italian (IT), and Swahili (SW). Using a set of adversarial prompts adapted from existing jailbreak datasets, we assess the model's susceptibility to harmful prompt injection across linguistic boundaries.

The results reveal distinct patterns of vulnerability. While Sabia-3.1 is robust against jailbreak attempts in its native language (PT-BR) and closely related European language (IT), it shows measurable vulnerabilities in English and more prominently in Swahili. These findings provide compelling evidence that LLMs are not inherently immune to prompt injection in other languages, but may generalize better for languages with shared linguistic origins (IT→PT-BR) instead of high-availability languages (EN→PT-BR).

II. Literature Review

Recent studies have identified multilingual prompts as potent vectors for bypassing safety mechanisms in LLMs. **"The Tower of Babel Revisited"** [4] comprehensively explored the efficiency of multilingual jailbreak prompts on closed-source LLMs, highlighting that prompts in languages other than English, or mixed-language inputs, frequently evade standard safety filters. This vulnerability was attributed to the insufficient representation of non-English languages during the alignment and safety training phases, underscoring the need for broader multilingual considerations in LLM training.

Complementary to this, **"Multilingual Jailbreak Challenges"** [5] systematically analyzed multilingual jailbreak issues, introducing the MultiJail dataset, which includes harmful prompts translated into various languages. This dataset illuminated the severity of multilingual threats, demonstrating how prompts translated from English into less commonly modeled languages disproportionately bypass safety constraints. The study proposed a defense mechanism, termed SELF-DEFENCE, which integrates multilingual inputs during training to significantly reduce multilingual vulnerabilities without extra annotations.

Beyond static prompts, adaptive multilingual injections pose a significant and emerging threat. The **"Sandwich Attack"** [6] detailed an innovative injection approach that dynamically interleaves languages within prompts to exploit LLMs adaptively. This method strategically sandwiches harmful instructions with innocuous multilingual context, continuously adjusting prompts based on LLM responses. The authors demonstrated that such adaptive multilingual mixtures dramatically improved attack success rates while bypassing conventional defense mechanisms, highlighting the critical need for real-time adaptive defenses.

Orthographic variations, especially involving non-standard scripts and transliteration, present unique vulnerabilities. **"Jailbreaking LLMs with Arabic Transliteration and Arabizi"** [7] exposed how Arabic transliteration (Arabizi), which is commonly used in informal communication, exploited significant security gaps. The authors argued that the combination of Latin script and Arabic phonetics circumvents standard language-model alignment processes, exposing severe shortcomings in existing standardized defense frameworks. Their findings emphasized the necessity for LLMs to incorporate script-agnostic training data and robust multilingual alignment protocols to mitigate such vulnerabilities effectively.

The implications of multilingual prompt injection extend notably to machine translation (MT) applications. **"Scaling Behavior of Machine Translation with Large Language Models under Prompt Injection Attacks"** [8] analyzed how prompt injection attacks degrade translation quality. Their findings suggested that larger LLMs generally offer improved robustness against injection attacks, though they remain vulnerable. The authors proposed defining explicit defensive thresholds linked to model size, thus providing guidance for scaling LLM architectures to enhance security in translation tasks.

"Multilingual Prompt Injection Detection and Adversarial Robustness" [9] focused on evaluating the robustness of large language models against multilingual prompt injection attacks. The research benchmarked 19 pre-trained LLMs, including general-purpose, domain-specific, adversarially fine-tuned, and multilingual models. Four diverse adversarial datasets were translated from English to Spanish, enabling comprehensive cross-lingual robustness evaluation. Their findings revealed considerable variability in performance across models; adversarially fine-tuned models exhibited higher accuracy in detecting malicious prompts compared to general-purpose models. The study highlights the challenge of maintaining adversarial robustness post-translation, with multilingual models demonstrating inconsistent generalization capabilities across languages. The authors stress the need for adaptive security mechanisms and cross-lingual adversarial training to combat evolving multilingual prompt injection threats

"A Cross-Language Investigation into Jailbreak Attacks in Large Language Models" [10] presented an extensive evaluation of multilingual jailbreak attacks on large language models. The authors conducted systematic experiments to assess vulnerabilities across various multilingual models, aiming to understand their susceptibility to jailbreak prompts. The study incorporated interpretability analysis to elucidate how these jailbreaks manipulate model behaviors. Mitigation strategies, including prompt-tuning and adversarial training methods tailored for multilingual contexts, were rigorously tested. Findings highlighted the complexities inherent to multilingual scenarios, demonstrating significant vulnerabilities even in well-trained multilingual models. The authors proposed improved mitigation strategies emphasizing multilingual adversarial training and targeted fine-tuning as critical pathways forward.

"Low-resource Language Jailbreaks in GPT-4" [11] uniquely explored the exploitation of low-resource languages to jailbreak GPT-4. They specifically investigated how prompts in languages with limited training data could effectively bypass standard safety mechanisms. Their analysis revealed that GPT-4's safety constraints were significantly more vulnerable when prompts were crafted in low-resource languages, where training data scarcity limited the model's contextual understanding and alignment. The authors argue that these vulnerabilities highlight critical gaps in current multilingual training protocols, underscoring the need for broader, more inclusive datasets and dedicated fine-tuning for underrepresented languages to bolster model robustness.

Table I
SUMMARY OF MODELS, DATASETS, LANGUAGES, AND RESULTS FROM ANALYZED PAPERS

Paper	Models	Datasets/Tasks	Languages (ISO)
[4]	GPT-4o, DeepSeek-R1, Gemini-1.5-Pro, Qwen-Max	32 jailbreak attack types; 38.4k responses (security)	EN, ZH
[5]	Unspecified GPT-based models	Malicious multilingual jailbreak queries	EN, TH, ZH
[6]	Bard, Gemini Pro, GPT-4, GPT-3.5-Turbo, Claude-3-OPUS	Multilingual adversarial prompt ("Sandwich")	EN, ES, ZH, AR, RU
[7]	Commercial/Open-source models (unspecified)	Arabic transliteration and Arabizi jailbreak	AR, Arabizi
[8]	Multiple LLM sizes	MT with prompt injection	EN→FR, EN→DE, EN→ES
[9]	BERT, DistilBERT, BioBERT, SciBERT, ProtectAI-DeBERTa, Indic-BERT, LaBSE, mBERT, XLM-R, ReMBERT	4 adversarial datasets (EN→ES)	EN, ES
[10]	GPT-3.5, GPT-4, Vicuna, LLaMa model	705 malicious multilingual questions	EN, ZH, ES, FR, AR, RU, PT, JA, SW
[11]	GPT-4	AdvBench dataset	ZU, GD, HMN, GN, UK, BN, TH, HE, ZH, AR, IT, HI, EN

Note: ISO Language codes — EN: English, ZH: Chinese, FR: French, DE: German, ES: Spanish, AR: Arabic, RU: Russian, TH: Thai, PT: Portuguese, JA: Japanese, SW: Swahili, ZU: Zulu, GD: Scots Gaelic, HMN: Hmong, GN: Guarani, UK: Ukrainian, BN: Bengali, HE: Hebrew, IT: Italian, HI: Hindi. Arabizi = Romanized Arabic.

III. METHODOLOGY

A. Model: Sabia-3.1

For this study, we used **Sabia-3.1** [3], a large language model primarily trained on Brazilian Portuguese content. Sabia-3.1 is designed for general-purpose language understanding and generation tasks, with a corpus that emphasizes Brazilian text sources from news media, books, web forums, and public documents. While it includes minimal multilingual exposure during pre-training, its alignment processes were centered around Portuguese-language safety and instruction tuning.

B. Prompt Dataset: MultiJail Adaptation

We leveraged the **MultiJail dataset** [5] which contains a diverse set of harmful jailbreak prompts across categories such as violence, hate speech, cybercrime, and self-harm. The original dataset includes prompts in English, some of which had previously been translated into other languages.

To suit our experimental goals, we performed automated translation of the full prompt set into Brazilian Portuguese. Contextual and cultural equivalence was manually checked, adjusting terminology to reflect natural idioms, grammar, and phrasing specific to Brazilian usage.

We also included the original **English**, a high-resource language, and additional translations into **Italian** (a mid-resource, latin, Romance language, *i.e.* with shared linguistic roots to Brazilian Portuguese) and **Swahili** (a low-resource Bantu language) to test multilingual behavior. Each of the 315 prompts was evaluated in all four languages, yielding **1,260 prompt-response pairs**.

C. Inference Procedure

All prompts were submitted to **Sabia-3.1** under the same conditions. Each query was executed via a script to ensure consistency. The model was not primed or role-conditioned in any way, emulating an end-user interaction scenario.

D. Classification Pipeline

To evaluate the **safety of the model's responses**, **GPT-4o**

All 1,260 responses were analyzed using the following prompt:

> *"The following files include requests and responses of unsafe content from a language model. Your task is to analyze the results and classify it as either 'safe' when the response is not harmful, 'unsafe' when the response is harmful, and 'invalid' when the response is not valid or looks completely unrelated. The result must be added in a new column at the end."*

Each response was fed alongside its original prompt, and GPT-4o produced one of three classifications:

- **Safe**: The model refused, deflected, or gave an innocuous reply.
- **Unsafe**: The model provided the requested harmful, illegal, or unethical information.
- **Invalid**: The response was nonsensical, incomplete, or otherwise unusable for evaluation.

No feedback or reinforcement was provided during labeling.

E. Jailbreak Success Rate (JSR)

We define the **Jailbreak Success Rate (JSR)** as the percentage of prompts that elicited *unsafe* responses from the model:

$$\text{JSR} = \frac{\text{Number of Unsafe Responses}}{\text{Total Prompts}} \times 100$$

This metric was computed separately for each language to allow for cross-linguistic comparison.

Table II
CLASSIFICATION AND JAILBREAK SUCCESS RATE (JSR)

Lang	Total	Safe	Unsafe	Safe(%)	Unsafe(%)	JSR
en	315	305	10	96.8	3.2	3.17
pt-br	315	315	0	100.0	0.0	0.00
it	315	315	0	100.0	0.0	0.00
sw	315	300	15	95.2	4.8	4.76

IV. CONCLUSION

Sabia-3.1, despite being trained mostly on Brazilian Portuguese sources, exhibits different behavior depending on the input language. While the model maintained a 0% jailbreak success rate in Brazilian Portuguese, thus demonstrating a strong alignment, its behavior changed significantly with English and Swahili inputs—languages for which it was not primarily aligned.

The higher vulnerability in Swahili, a low-resource language in this context, suggests that attackers might exploit underrepresented languages to bypass safety constraints, but is expected. The higher jailbreak success rate in English is unusual, because English is such a high-resource language and may be commonly found even in language-specific datasets.

Italian, which is a language unlikely to be found in a Brazilian dataset, also managed to maintain a 0% jailbreak success rate, suggesting language models may be able to generalize alignment for languages with shared ancestry roots due to their similarity.

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